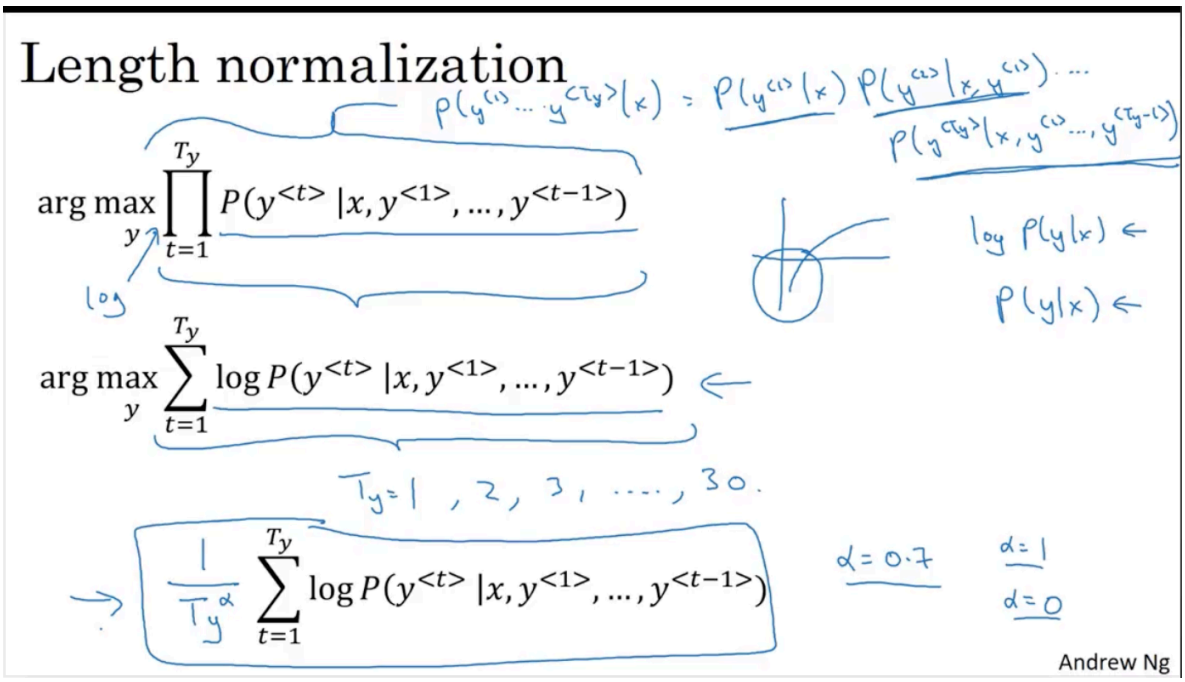
C5W3L04 Refining Beam Search

Refining Beam search may help us to make it better

Length Normalization in Beam Search

The basic scoring function in beam search has an implicit bias toward shorter sequences.
The reason is that the probability of an entire sequence is calculated as the product of individual token probabilities. Since each probability is between 0 and 1, adding more tokens generally decreases the overall sequence probability.
In practice, we work with log-probabilities, which means summing individual token log-probabilities. Longer sequences tend to accumulate more negative values.
We want to mitigate this bias because a longer sequence isn't necessarily worse than a shorter one.
That's why we introduce **length normalization**, modifying the scoring function to consider the average log-probability per token rather than the total log-probability.
This allows us to compare sequences based on how probable their tokens are on average, rather than systematically favoring shorter sequences.
Importantly, length normalization doesn't blindly encourage longer sequences. It simply reduces the disadvantage that longer sequences have under the original scoring function.

One final conceptual distinction worth remembering:
Length normalization is a modification of the search/scoring procedure, not the language model itself. The model still predicts exactly the same conditional probabilities. We only change how beam search evaluates candidate sequences.



Don’t choose B to be very big, because it makes computation slower

Beam search discussion

Beam width B?

Unlike exact search algorithms like BFS (Breadth First Search) or DFS (Depth First Search), Beam Search runs faster but is not guaranteed to find exact maximum for $\arg\max_y P(y|x)$.